

## **A BRIEF SUMMARY OF THE IMPLEMENTATION.**

Development Phase/reflection phase:

This database is designed with the schema of an Airbnb-like platform. It is called `airbnb_company`, and it has 21 interrelated tables. The database is well organized in respect to user, properties, bookings, payments, reviews, and so on. The schema contains the multiples tables to manage the several features of the platform such as user administration, property management, booking transactions, payment transactions, review management, amenities and system administration.

The user table assumes the role of a central entity, user being the most important refers to the host assistance admin. Host and Guest are stored in the Host and Guest tables respectively with additional details like bank account and payment options. Properties are stored in the Property table which has a relationship with Location table for address and Amenity table for property features. Booking is maintained in booking table which relates properties and guests while payments are done through Payment table. Reviews are stored in Review table allowing users to rate and comment on their experiences while utilizing our service.

Additional tables for this database includes Commission, Payment, Message, Social Media, Dispute, Notification, Income and Language provide the specific functionalities like giving information for the property availability, commission, tracking, payout, messaging, social media interaction, dispute resolution, notifications income tracking, and language preferences can be accessed conveniently.

The architecture of this database design provides an integrated and scalable solution in the management of the intricate interaction and data flow as in any schema of the Airbnb platform. However, this supports user roles, property listings, bookings, payments, reviews, and administrative-related tasks while safeguarding data integrity by means of well-defined relationships and constraints.

Creating the database schema named as airbnb\_company

```
CREATE DATABASE IF NOT EXISTS  
airbnb_company;
```

The database comprises 21 interconnected tables, each meticulously crafted to handle specific entities and relationships with the Airbnb ecosystem platform.

## User:

This entity can store user details including name, email, phone and type (Host, Guest, Admin).

Use the SQL statement to CREATE TABLE to define the structure of the User table.

```
CREATE TABLE airbnb_company.User (  
  id INT PRIMARY KEY AUTO_INCREMENT,  
  name VARCHAR(255) NOT NULL,  
  email VARCHAR(255) UNIQUE NOT NULL,  
  phone VARCHAR(15),  
  user_type ENUM('Host', 'Guest', 'Admin') NOT NULL  
);
```

Detailed information of user type (Host, Guest, Admin) can be inserted into the table using INSERT INTO statements.

```
INSERT INTO airbnb_company.User (name, email, phone, user_type)  
VALUES  
(  
  'John Doe', 'john.doe@gmail.com', '1234567890', 'Host'),  
  ('Jane Smith', 'jane.smith@hotmail.com', '0987654321', 'Guest'),  
  ('Alice Johnson', 'alice.johnson@gmail.com', '1122334455', 'Admin'),  
  ('Bob Brown', 'bob.brown@gmail.com', '2233445566', 'Host'),  
  ('Charlie Davis', 'charlie.davis@hotmail.com', '3344556677', 'Guest'),  
  ('Diana Evans', 'diana.evans@hotmail.com', '4455667788', 'Admin'),  
  ('Ethan Green', 'ethan.green@yahoo.com', '5566778899', 'Host'),  
  ('Fiona Harris', 'fiona.harris@gmail.com', '6677889900', 'Guest'),  
  ('George Clark', 'george.clark@hotmail.com', '7788990011', 'Admin'),  
  ('Hannah Lewis', 'hannah.lewis@yahoo.com', '8899001122', 'Host'),  
  ('Ian Walker', 'ian.walker@gmail.com', '9900112233', 'Guest'),  
  ('Jessica Hall', 'jessica.hall@hotmail.com', '0011223344', 'Admin'),  
  ('Kevin Allen', 'kevin.allen@yahoo.com', '1122334455', 'Host'),  
  ('Laura Young', 'laura.young@hotmail.com', '2233445566', 'Guest'),  
  ('Michael King', 'michael.king@yahoo.com', '3344556677', 'Admin'),  
  ('Nina Wright', 'nina.wright@gmail.com', '4455667788', 'Host'),  
  ('Oscar Scott', 'oscar.scott@yahoo.com', '5566778899', 'Guest'),  
  ('Paula Turner', 'paula.turner@hotmail.com', '6677889900', 'Admin'),  
  ('Quincy Adams', 'quincy.adams@gmail.com', '7788990011', 'Host'),  
  ('Rachel White', 'rachel.white@hotmail.com', '8899001122', 'Guest');
```

| id   | name          | email                     | phone      | user_type |
|------|---------------|---------------------------|------------|-----------|
| 1    | John Doe      | john.doe@gmail.com        | 1234567890 | Host      |
| 2    | Jane Smith    | jane.smith@hotmail.com    | 0987654321 | Guest     |
| 3    | Alice Johnson | alice.johnson@gmail.com   | 1122334455 | Admin     |
| 4    | Bob Brown     | bob.brown@gmail.com       | 2233445566 | Host      |
| 5    | Charlie Davis | charlie.davis@hotmail.com | 3344556677 | Guest     |
| 6    | Diana Evans   | diana.evans@hotmail.com   | 4455667788 | Admin     |
| 7    | Ethan Green   | ethan.green@yahoo.com     | 5566778899 | Host      |
| 8    | Fiona Harris  | fiona.harris@gmail.com    | 6677889900 | Guest     |
| 9    | George Clark  | george.clark@hotmail.com  | 7788990011 | Admin     |
| 10   | Hannah Lewis  | hannah.lewis@yahoo.com    | 8899001122 | Host      |
| 11   | Ian Walker    | ian.walker@gmail.com      | 9900112233 | Guest     |
| 12   | Jessica Hall  | jessica.hall@hotmail.com  | 0011223344 | Admin     |
| 13   | Kevin Allen   | kevin.allen@yahoo.com     | 1122334455 | Host      |
| 14   | Laura Young   | laura.young@hotmail.com   | 2233445566 | Guest     |
| 15   | Michael King  | michael.king@yahoo.com    | 3344556677 | Admin     |
| 16   | Nina Wright   | nina.wright@gmail.com     | 4455667788 | Host      |
| 17   | Oscar Scott   | oscar.scott@yahoo.com     | 5566778899 | Guest     |
| 18   | Paula Turner  | paula.turner@hotmail.com  | 6677889900 | Admin     |
| 19   | Quincy Adams  | quincy.adams@gmail.com    | 7788990011 | Host      |
| 20   | Rachel White  | rachel.white@hotmail.com  | 8899001122 | Guest     |
| NULL | NULL          | NULL                      | NULL       | NULL      |

User: Test case: Calculate the number of admin users:  
This query aims to **count the number of admin users** (totals\_admins) for each user type (user\_type) in the **airbnb\_company** database. It helps in understanding how many users of each type have admin privileges.

|  | user_type | total_admins |
|--|-----------|--------------|
|  | Host      | 0            |
|  | Guest     | 0            |
|  | Admin     | 6            |
|  |           |              |

```
1 • USE airbnb_company;
2
3 • SELECT
4     u.user_type,
5     COUNT(a.id) AS total_admins
6 FROM
7     User u
8 LEFT JOIN Admin a ON u.id = a.user_id
9 GROUP BY u.user_type;
```

- USE Clause: Use the USE Clause to retrieve the data from **airbnb\_company** database.
- SELECT Clause: u.user\_type: This selects the user\_type column from the User table. It represents the type of user ( e.g., “gu est”, “ host”, “admin”),  
COUNT(a.id) AS total\_admins: This counts the number of admin records (a.id) associated with each user\_type. If a user does not have an admin record. It will count as 0 due to the LEFT JOIN, aliasing it as ‘total\_admins’.
- FROM Clause: User u: Specifies the ‘User” table and aliases it as ‘u’.
- LEFT JOIN Admin a ON u.id = a.user\_id: Left join can ensure that all rows from User table are included in the result, even there is no matching row in the Admin table, as well as ON u.id = a.user\_id can link the User table to ( u.id) to Admin table (a.user\_id) based on the user’s ID. If a user does not have an admin record, the Admin table columns will show as NULL.
- GROUP BY u.user\_type: This groups the results by user\_type. For each unique user\_type, the query calculates the total number of admins( total\_admins).

Host:

This entity can store the host-specific details linked to a user, including bank account information.

Use the SQL statement to CREATE TABLE to define the structure of the Host table.

```
• CREATE TABLE airbnb_company.Host (  
    id INT PRIMARY KEY AUTO_INCREMENT,  
    user_id INT NOT NULL,  
    bank_account_details VARCHAR(255) NOT NULL,  
    FOREIGN KEY (user_id) REFERENCES User(id)  
);
```

Inserting the details of host-specific linked to a user in the table using INSERT INTO statements.

```
• INSERT INTO airbnb_company.Host (user_id, bank_account_details)  
VALUES  
(1, 'Bank of America, Account Number: 123456789'),  
(2, 'Chase, Account Number: 987654321'),  
(3, 'Danske bank, Account Number: 456789123'),  
(4, 'Citibank, Account Number: 321654987'),  
(5, 'US Bank, Account Number: 789123456'),  
(6, 'Capital One, Account Number: 654987321'),  
(7, 'TD Bank, Account Number: 123789456'),  
(8, 'PNC Bank, Account Number: 987321654'),  
(9, 'HSBC, Account Number: 456123789'),  
(10, 'Ally Bank, Account Number: 321987654'),  
(11, 'Discover Bank, Account Number: 789456123'),  
(12, 'Barclays, Account Number: 654321987'),  
(13, 'Santander Bank, Account Number: 123456987'),  
(14, 'Regions Bank, Account Number: 987654123'),  
(15, 'Fifth Third Bank, Account Number: 456789321'),  
(16, 'jysk Bank, Account Number: 321456987'),  
(17, 'SunTrust, Account Number: 789123654'),  
(18, 'Novo Banco, Account Number: 654789321'),  
(19, 'Charles Schwab Bank, Account Number: 123987456'),  
(20, 'Synchrony Bank, Account Number: 987123654');
```

| id   | user_id | bank_account_details                           |
|------|---------|------------------------------------------------|
| 1    | 1       | Bank of America, Account Number: 123456789     |
| 2    | 2       | Chase, Account Number: 987654321               |
| 3    | 3       | Danske bank, Account Number: 456789123         |
| 4    | 4       | Citibank, Account Number: 321654987            |
| 5    | 5       | US Bank, Account Number: 789123456             |
| 6    | 6       | Capital One, Account Number: 654987321         |
| 7    | 7       | TD Bank, Account Number: 123789456             |
| 8    | 8       | PNC Bank, Account Number: 987321654            |
| 9    | 9       | HSBC, Account Number: 456123789                |
| 10   | 10      | Ally Bank, Account Number: 321987654           |
| 11   | 11      | Discover Bank, Account Number: 789456123       |
| 12   | 12      | Barclays, Account Number: 654321987            |
| 13   | 13      | Santander Bank, Account Number: 123456987      |
| 14   | 14      | Regions Bank, Account Number: 987654123        |
| 15   | 15      | Fifth Third Bank, Account Number: 456789321    |
| 16   | 16      | jysk Bank, Account Number: 321456987           |
| 17   | 17      | SunTrust, Account Number: 789123654            |
| 18   | 18      | Novo Banco, Account Number: 654789321          |
| 19   | 19      | Charles Schwab Bank, Account Number: 123987456 |
| 20   | 20      | Synchrony Bank, Account Number: 987123654      |
| NULL | NULL    | NULL                                           |

Guest:

This entity can store guest-specific details linked to a user, including payment method.

Use the SQL statement to CREATE TABLE to define the structure of the Guest table.

```
• CREATE TABLE airbnb_company.Guest (  
  id INT PRIMARY KEY AUTO_INCREMENT,  
  user_id INT NOT NULL,  
  payment_method VARCHAR(255) NOT NULL,  
  FOREIGN KEY (user_id) REFERENCES User(id)  
);
```

Inserting the details of guest-specific linked to a user in the table using INSERT INTO statements.

```
• INSERT INTO airbnb_company.Guest (user_id, payment_method)  
VALUES  
(2, 'Credit Card'),  
(5, 'PayPal'),  
(8, 'Credit Card'),  
(11, 'PayPal'),  
(14, 'Credit Card'),  
(17, 'PayPal'),  
(20, 'Credit Card'),  
(1, 'PayPal'),  
(4, 'Credit Card'),  
(7, 'PayPal'),  
(10, 'Credit Card'),  
(13, 'PayPal'),  
(16, 'Credit Card'),  
(19, 'PayPal'),  
(3, 'Credit Card'),  
(6, 'PayPal'),  
(9, 'Credit Card'),  
(12, 'PayPal'),  
(15, 'Credit Card'),  
(18, 'PayPal');
```

|  | id   | user_id | payment_meth... |
|--|------|---------|-----------------|
|  | 1    | 2       | Credit Card     |
|  | 2    | 5       | PayPal          |
|  | 3    | 8       | Credit Card     |
|  | 4    | 11      | PayPal          |
|  | 5    | 14      | Credit Card     |
|  | 6    | 17      | PayPal          |
|  | 7    | 20      | Credit Card     |
|  | 8    | 1       | PayPal          |
|  | 9    | 4       | Credit Card     |
|  | 10   | 7       | PayPal          |
|  | 11   | 10      | Credit Card     |
|  | 12   | 13      | PayPal          |
|  | 13   | 16      | Credit Card     |
|  | 14   | 19      | PayPal          |
|  | 15   | 3       | Credit Card     |
|  | 16   | 6       | PayPal          |
|  | 17   | 9       | Credit Card     |
|  | 18   | 12      | PayPal          |
|  | 19   | 15      | Credit Card     |
|  | 20   | 18      | PayPal          |
|  | NULL | NULL    | NULL            |

Location:

This entity store property location details like country, city, street, and postcode.

Use the SQL statement to CREATE TABLE to define the structure of the Location table.

```
CREATE TABLE airbnb_company.Location (  
  id INT PRIMARY KEY AUTO_INCREMENT,  
  country VARCHAR(255) NOT NULL,  
  city VARCHAR(255) NOT NULL,  
  street VARCHAR(255) NOT NULL,  
  post_code VARCHAR(255) NOT NULL  
);
```

Inserting the location details of property in the table using INSERT INTO statements.

```
INSERT INTO airbnb_company.Location (country, city, street, post_code)  
VALUES  
( 'USA', 'New York', '5th Ave', '10001'),  
( 'USA', 'Los Angeles', 'Sunset Blvd', '90001'),  
( 'UK', 'London', 'Oxford St', 'W1D 1BS'),  
( 'France', 'Paris', 'Champs-Élysées', '75008'),  
( 'Germany', 'Berlin', 'Unter den Linden', '10117'),  
( 'Italy', 'Rome', 'Via del Corso', '00186'),  
( 'Spain', 'Madrid', 'Gran Vía', '28013'),  
( 'Canada', 'Toronto', 'Yonge St', 'M5B 2H1'),  
( 'Australia', 'Sydney', 'George St', '2000'),  
( 'Japan', 'Tokyo', 'Ginza', '104-0061'),  
( 'China', 'Beijing', 'Wangfujing', '100006'),  
( 'Brazil', 'Rio de Janeiro', 'Copacabana', '22070-012'),  
( 'India', 'Mumbai', 'Marine Drive', '400020'),  
( 'South Africa', 'Cape Town', 'Long St', '8001'),  
( 'Russia', 'Moscow', 'Tverskaya St', '125009'),  
( 'Mexico', 'Mexico City', 'Paseo de la Reforma', '06500'),  
( 'Argentina', 'Buenos Aires', 'Avenida 9 de Julio', 'C1043'),  
( 'Egypt', 'Cairo', 'Talaat Harb St', '11511'),  
( 'Turkey', 'Istanbul', 'İstiklal Caddesi', '34433'),  
( 'Thailand', 'Bangkok', 'Sukhumvit Rd', '10110');
```

|  | id   | country    | city          | street             | post_code |
|--|------|------------|---------------|--------------------|-----------|
|  | 1    | USA        | New York      | 5th Ave            | 10001     |
|  | 2    | USA        | Los Angeles   | Sunset Blvd        | 90001     |
|  | 3    | UK         | London        | Oxford St          | W1D 1BS   |
|  | 4    | France     | Paris         | Champs-Élysées     | 75008     |
|  | 5    | Germany    | Berlin        | Unter den Linden   | 10117     |
|  | 6    | Italy      | Rome          | Via del Corso      | 00186     |
|  | 7    | Spain      | Madrid        | Gran Vía           | 28013     |
|  | 8    | Canada     | Toronto       | Yonge St           | M5B 2H1   |
|  | 9    | Australia  | Sydney        | George St          | 2000      |
|  | 10   | Japan      | Tokyo         | Ginza              | 104-0061  |
|  | 11   | China      | Beijing       | Wangfujing         | 100006    |
|  | 12   | Brazil     | Rio de Jan... | Copacabana         | 22070-012 |
|  | 13   | India      | Mumbai        | Marine Drive       | 400020    |
|  | 14   | South A... | Cape Town     | Long St            | 8001      |
|  | 15   | Russia     | Moscow        | Tverskaya St       | 125009    |
|  | 16   | Mexico     | Mexico City   | Paseo de la Ref... | 06500     |
|  | 17   | Argentina  | Buenos Aires  | Avenida 9 de Julio | C1043     |
|  | 18   | Egypt      | Cairo         | Talaat Harb St     | 11511     |
|  | 19   | Turkey     | Istanbul      | İstiklal Caddesi   | 34433     |
|  | 20   | Thailand   | Bangkok       | Sukhumvit Rd       | 10110     |
|  | NULL | NULL       | NULL          | NULL               | NULL      |
|  |      |            |               |                    |           |

Location: Test case :Count the total number of properties located on each country.  
This query retrieves the count the total number of properties located on each country based on the data stored in two tables: Location and Property, which helps to find out the numbers of properties on each country stored on database.

```
USE airbnb_company;

SELECT
  l.country,
  COUNT(p.id) AS total_properties
FROM
  Location l
JOIN
  Property p
  ON l.id = p.location_id
GROUP BY
  l.country
ORDER BY
  total_properties DESC;
```

|  | country    | total_properti... |
|--|------------|-------------------|
|  | USA        | 2                 |
|  | UK         | 1                 |
|  | France     | 1                 |
|  | Germany    | 1                 |
|  | Italy      | 1                 |
|  | Spain      | 1                 |
|  | Canada     | 1                 |
|  | Australia  | 1                 |
|  | Japan      | 1                 |
|  | China      | 1                 |
|  | Brazil     | 1                 |
|  | India      | 1                 |
|  | South A... | 1                 |
|  | Russia     | 1                 |
|  | Mexico     | 1                 |
|  | Argentina  | 1                 |
|  | Egypt      | 1                 |
|  | Turkey     | 1                 |
|  | Thailand   | 1                 |

USE Clause: use the Use Clause to retrieve the data from **airbnb\_company** database.

SELECT Clause: l.country: This selects the country columns form the Location table, COUNT(p.id) AS total\_properties: This counts the number of property IDS (p.id) and labels the result as total\_properties. This gives us the total number of properties in each country.

FROM Clause: Location l : Specifies the “Location” table and aliases it as ‘l’.

JOIN Clause: Property p: This specifies the Property table from the airbnb\_company database and assigns it an alias p,

ON l.id = p.location\_id: This joins the Location and Property tables where the id column in the Location table matches the location\_id column in the Property table. This ensures that we are only considering properties that have a valid location.

GROUP BY Clause: l.country: This groups the results by the country column. This means that the query will aggregate the data (in this case, count the properties) for each unique country.

ORDER BY Clause: total\_properties DESC: This orders the results by the total\_properties column in descending order. This means that the country with the highest number of properties will appear first in the result set.

## Property:

This entity can store property details including host, name, description, location and price per night.

Use the SQL statement to CREATE TABLE to define the structure of the Property table.

```
CREATE TABLE airbnb_company.Property (  
  id INT PRIMARY KEY AUTO_INCREMENT,  
  host_id INT NOT NULL,  
  name VARCHAR(255) NOT NULL,  
  description TEXT,  
  location_id INT NOT NULL,  
  price_per_night DECIMAL(10, 2) NOT NULL,  
  FOREIGN KEY (host_id) REFERENCES Host(id),  
  FOREIGN KEY (location_id) REFERENCES Location(id)  
);
```

Detailed description and name of property linked to the host and location with price per night can be inserted in the table using INSERT INTO statements.

```
INSERT INTO airbnb_company.Property (host_id, name, description, location_id, price_per_night)  
VALUES  
(1, 'Cozy Apartment', 'A cozy apartment in the heart of the city.', 1, 100.00),  
(2, 'Luxury Villa', 'A luxurious villa with a private pool.', 2, 500.00),  
(3, 'City Loft', 'A modern loft with stunning city views.', 3, 200.00),  
(4, 'Beach House', 'A beautiful house right on the beach.', 4, 300.00),  
(5, 'Mountain Cabin', 'A rustic cabin in the mountains.', 5, 150.00),  
(6, 'Downtown Condo', 'A stylish condo in the downtown area.', 6, 250.00),  
(7, 'Country Cottage', 'A charming cottage in the countryside.', 7, 120.00),  
(8, 'Lakefront Retreat', 'A peaceful retreat by the lake.', 8, 180.00),  
(9, 'Urban Studio', 'A compact studio in the urban jungle.', 9, 90.00),  
(10, 'Historic Townhouse', 'A historic townhouse with modern amenities.', 10, 220.00),  
(11, 'Ski Chalet', 'A cozy chalet near the ski slopes.', 11, 400.00),  
(12, 'Riverside Bungalow', 'A quaint bungalow by the river.', 12, 130.00),  
(13, 'Penthouse Suite', 'A luxurious penthouse with panoramic views.', 13, 600.00),  
(14, 'Garden Apartment', 'A bright apartment with a private garden.', 14, 160.00),  
(15, 'Seaside Villa', 'A stunning villa with ocean views.', 15, 450.00),  
(16, 'Forest Retreat', 'A secluded retreat in the forest.', 16, 140.00),  
(17, 'Modern Apartment', 'A sleek and modern apartment.', 17, 170.00),  
(18, 'Traditional House', 'A traditional house with a cultural touch.', 18, 190.00),  
(19, 'Luxury Apartment', 'A high-end apartment with all amenities.', 19, 350.00),  
(20, 'City View Apartment', 'An apartment with a great view of the city.', 20, 210.00);
```

| id   | host_id | name                | description                                 | location_id | price_per_ni... |
|------|---------|---------------------|---------------------------------------------|-------------|-----------------|
| 1    | 1       | Cozy Apartment      | A cozy apartment in the heart of the city.  | 1           | 100.00          |
| 2    | 2       | Luxury Villa        | A luxurious villa with a private pool.      | 2           | 500.00          |
| 3    | 3       | City Loft           | A modern loft with stunning city views.     | 3           | 200.00          |
| 4    | 4       | Beach House         | A beautiful house right on the beach.       | 4           | 300.00          |
| 5    | 5       | Mountain Cabin      | A rustic cabin in the mountains.            | 5           | 150.00          |
| 6    | 6       | Downtown Condo      | A stylish condo in the downtown area.       | 6           | 250.00          |
| 7    | 7       | Country Cottage     | A charming cottage in the countryside.      | 7           | 120.00          |
| 8    | 8       | Lakefront Retreat   | A peaceful retreat by the lake.             | 8           | 180.00          |
| 9    | 9       | Urban Studio        | A compact studio in the urban jungle.       | 9           | 90.00           |
| 10   | 10      | Historic Townhouse  | A historic townhouse with modern amenities. | 10          | 220.00          |
| 11   | 11      | Ski Chalet          | A cozy chalet near the ski slopes.          | 11          | 400.00          |
| 12   | 12      | Riverside Bungalow  | A quaint bungalow by the river.             | 12          | 130.00          |
| 13   | 13      | Penthouse Suite     | A luxurious penthouse with panoramic views. | 13          | 600.00          |
| 14   | 14      | Garden Apartment    | A bright apartment with a private garden.   | 14          | 160.00          |
| 15   | 15      | Seaside Villa       | A stunning villa with ocean views.          | 15          | 450.00          |
| 16   | 16      | Forest Retreat      | A secluded retreat in the forest.           | 16          | 140.00          |
| 17   | 17      | Modern Apartment    | A sleek and modern apartment.               | 17          | 170.00          |
| 18   | 18      | Traditional House   | A traditional house with a cultural touch.  | 18          | 190.00          |
| 19   | 19      | Luxury Apartment    | A high-end apartment with all amenities.    | 19          | 350.00          |
| 20   | 20      | City View Apartment | An apartment with a great view of the city. | 20          | 210.00          |
| NULL | NULL    | NULL                | NULL                                        | NULL        | NULL            |

Booking:  
The entity can store booking details including property, guest, dates, total price and status.

Use the SQL statement to CREATE TABLE to define the structure of the Booking table.

```
CREATE TABLE airbnb_company.Booking (  
  id INT PRIMARY KEY AUTO_INCREMENT,  
  property_id INT NOT NULL,  
  guest_id INT NOT NULL,  
  check_in_date DATE NOT NULL,  
  check_out_date DATE NOT NULL,  
  total_price DECIMAL(10,2) NOT NULL,  
  status ENUM('Pending', 'Confirmed', 'Cancelled', 'Completed') NOT NULL DEFAULT 'Pending',  
  FOREIGN KEY (property_id) REFERENCES Property(id),  
  FOREIGN KEY (guest_id) REFERENCES Guest(id)  
);
```

Inserting the booking detailed with check-in, check-out with booking status and total price of booking can be showed in the table using INSERT INTO statements.

```
INSERT INTO airbnb_company.Booking (property_id, guest_id, check_in_date, check_out_date, total_price, status)  
VALUES  
(1, 1, '2023-10-01', '2023-10-05', 400.00, 'Completed'),  
(2, 2, '2023-10-02', '2023-10-07', 2500.00, 'Confirmed'),  
(3, 3, '2023-10-03', '2023-10-06', 600.00, 'Pending'),  
(4, 4, '2023-10-04', '2023-10-09', 1500.00, 'Confirmed'),  
(5, 5, '2023-10-05', '2023-10-10', 750.00, 'Completed'),  
(6, 6, '2023-10-06', '2023-10-11', 1250.00, 'Confirmed'),  
(7, 7, '2023-10-07', '2023-10-12', 600.00, 'Pending'),  
(8, 8, '2023-10-08', '2023-10-13', 900.00, 'Confirmed'),  
(9, 9, '2023-10-09', '2023-10-14', 450.00, 'Completed'),  
(10, 10, '2023-10-10', '2023-10-15', 1100.00, 'Confirmed'),  
(11, 11, '2023-10-11', '2023-10-16', 2000.00, 'Pending'),  
(12, 12, '2023-10-12', '2023-10-17', 650.00, 'Confirmed'),  
(13, 13, '2023-10-13', '2023-10-18', 3000.00, 'Completed'),  
(14, 14, '2023-10-14', '2023-10-19', 800.00, 'Confirmed'),  
(15, 15, '2023-10-15', '2023-10-20', 2250.00, 'Pending'),  
(16, 16, '2023-10-16', '2023-10-21', 700.00, 'Confirmed'),  
(17, 17, '2023-10-17', '2023-10-22', 850.00, 'Completed'),  
(18, 18, '2023-10-18', '2023-10-23', 950.00, 'Confirmed'),  
(19, 19, '2023-10-19', '2023-10-24', 1750.00, 'Pending'),  
(20, 20, '2023-10-20', '2023-10-25', 1050.00, 'Confirmed');
```

| id   | property_id | guest_id | check_in_da... | check_out_da... | total_pri... | status    |
|------|-------------|----------|----------------|-----------------|--------------|-----------|
| 1    | 1           | 1        | 2023-10-01     | 2023-10-05      | 400.00       | Completed |
| 2    | 2           | 2        | 2023-10-02     | 2023-10-07      | 2500.00      | Confirmed |
| 3    | 3           | 3        | 2023-10-03     | 2023-10-06      | 600.00       | Pending   |
| 4    | 4           | 4        | 2023-10-04     | 2023-10-09      | 1500.00      | Confirmed |
| 5    | 5           | 5        | 2023-10-05     | 2023-10-10      | 750.00       | Completed |
| 6    | 6           | 6        | 2023-10-06     | 2023-10-11      | 1250.00      | Confirmed |
| 7    | 7           | 7        | 2023-10-07     | 2023-10-12      | 600.00       | Pending   |
| 8    | 8           | 8        | 2023-10-08     | 2023-10-13      | 900.00       | Confirmed |
| 9    | 9           | 9        | 2023-10-09     | 2023-10-14      | 450.00       | Completed |
| 10   | 10          | 10       | 2023-10-10     | 2023-10-15      | 1100.00      | Confirmed |
| 11   | 11          | 11       | 2023-10-11     | 2023-10-16      | 2000.00      | Pending   |
| 12   | 12          | 12       | 2023-10-12     | 2023-10-17      | 650.00       | Confirmed |
| 13   | 13          | 13       | 2023-10-13     | 2023-10-18      | 3000.00      | Completed |
| 14   | 14          | 14       | 2023-10-14     | 2023-10-19      | 800.00       | Confirmed |
| 15   | 15          | 15       | 2023-10-15     | 2023-10-20      | 2250.00      | Pending   |
| 16   | 16          | 16       | 2023-10-16     | 2023-10-21      | 700.00       | Confirmed |
| 17   | 17          | 17       | 2023-10-17     | 2023-10-22      | 850.00       | Completed |
| 18   | 18          | 18       | 2023-10-18     | 2023-10-23      | 950.00       | Confirmed |
| 19   | 19          | 19       | 2023-10-19     | 2023-10-24      | 1750.00      | Pending   |
| 20   | 20          | 20       | 2023-10-20     | 2023-10-25      | 1050.00      | Confirmed |
| NULL | NULL        | NULL     | NULL           | NULL            | NULL         | NULL      |

Payment:

This entity can store payment details linked to a booking, including amount and payment date.

Use the SQL statement to CREATE TABLE to define the structure of the Payment table.

```
CREATE TABLE airbnb_company.Payment (  
  id INT PRIMARY KEY AUTO_INCREMENT,  
  booking_id INT NOT NULL,  
  amount DECIMAL(10, 2) NOT NULL,  
  payment_date DATE NOT NULL,  
  payment_method VARCHAR(255) NOT NULL,  
  FOREIGN KEY (booking_id) REFERENCES Booking(id)  
);
```

Detailed information about payment linked to booking with amount can be inserted in the table using INSERT INTO statements.

```
INSERT INTO airbnb_company.Payment (booking_id, amount, payment_date, payment_method)  
VALUES  
(1, 400.00, '2023-10-01', 'Credit Card'),  
(2, 2500.00, '2023-10-02', 'PayPal'),  
(3, 600.00, '2023-10-03', 'Credit Card'),  
(4, 1500.00, '2023-10-04', 'PayPal'),  
(5, 750.00, '2023-10-05', 'Credit Card'),  
(6, 1250.00, '2023-10-06', 'PayPal'),  
(7, 600.00, '2023-10-07', 'Credit Card'),  
(8, 900.00, '2023-10-08', 'PayPal'),  
(9, 450.00, '2023-10-09', 'Credit Card'),  
(10, 1100.00, '2023-10-10', 'PayPal'),  
(11, 2000.00, '2023-10-11', 'Credit Card'),  
(12, 650.00, '2023-10-12', 'PayPal'),  
(13, 3000.00, '2023-10-13', 'Credit Card'),  
(14, 800.00, '2023-10-14', 'PayPal'),  
(15, 2250.00, '2023-10-15', 'Credit Card'),  
(16, 700.00, '2023-10-16', 'PayPal'),  
(17, 850.00, '2023-10-17', 'Credit Card'),  
(18, 950.00, '2023-10-18', 'PayPal'),  
(19, 1750.00, '2023-10-19', 'Credit Card'),  
(20, 1050.00, '2023-10-20', 'PayPal');
```

|  | id   | booking_id | amount  | payment_date | payment_meth... |
|--|------|------------|---------|--------------|-----------------|
|  | 1    | 1          | 400.00  | 2023-10-01   | Credit Card     |
|  | 2    | 2          | 2500.00 | 2023-10-02   | PayPal          |
|  | 3    | 3          | 600.00  | 2023-10-03   | Credit Card     |
|  | 4    | 4          | 1500.00 | 2023-10-04   | PayPal          |
|  | 5    | 5          | 750.00  | 2023-10-05   | Credit Card     |
|  | 6    | 6          | 1250.00 | 2023-10-06   | PayPal          |
|  | 7    | 7          | 600.00  | 2023-10-07   | Credit Card     |
|  | 8    | 8          | 900.00  | 2023-10-08   | PayPal          |
|  | 9    | 9          | 450.00  | 2023-10-09   | Credit Card     |
|  | 10   | 10         | 1100.00 | 2023-10-10   | PayPal          |
|  | 11   | 11         | 2000.00 | 2023-10-11   | Credit Card     |
|  | 12   | 12         | 650.00  | 2023-10-12   | PayPal          |
|  | 13   | 13         | 3000.00 | 2023-10-13   | Credit Card     |
|  | 14   | 14         | 800.00  | 2023-10-14   | PayPal          |
|  | 15   | 15         | 2250.00 | 2023-10-15   | Credit Card     |
|  | 16   | 16         | 700.00  | 2023-10-16   | PayPal          |
|  | 17   | 17         | 850.00  | 2023-10-17   | Credit Card     |
|  | 18   | 18         | 950.00  | 2023-10-18   | PayPal          |
|  | 19   | 19         | 1750.00 | 2023-10-19   | Credit Card     |
|  | 20   | 20         | 1050.00 | 2023-10-20   | PayPal          |
|  | NULL | NULL       | NULL    | NULL         | NULL            |

## Review:

This entity can store reviews linked to bookings, including ratings, comments, and reviewer/reviewee details.

Use the SQL statement to CREATE TABLE to define the structure of the Review table.

```
CREATE TABLE airbnb_company.Review (  
  id INT PRIMARY KEY AUTO_INCREMENT,  
  booking_id INT NOT NULL,  
  reviewer_id INT NOT NULL,  
  reviewee_id INT NOT NULL,  
  rating INT CHECK (rating BETWEEN 1 AND 5),  
  comment TEXT,  
  review_date DATE NOT NULL,  
  FOREIGN KEY (booking_id) REFERENCES Booking(id),  
  FOREIGN KEY (reviewer_id) REFERENCES User(id),  
  FOREIGN KEY (reviewee_id) REFERENCES User(id)  
);
```

Detailed reviews linked to booking and user's rating and comment can be inserted using INSERT INTO statements.

```
INSERT INTO airbnb_company.Review (booking_id, reviewer_id, reviewee_id, rating, comment, review_date)  
VALUES  
(1, 1, 1, 5, 'Great stay!', '2023-10-05'),  
(2, 2, 2, 4, 'Loved the place!', '2023-10-07'),  
(3, 3, 3, 3, 'It was okay.', '2023-10-06'),  
(4, 4, 4, 5, 'Amazing experience!', '2023-10-09'),  
(5, 5, 5, 4, 'Very comfortable.', '2023-10-10'),  
(6, 6, 6, 5, 'Perfect location!', '2023-10-11'),  
(7, 7, 7, 3, 'Could be better.', '2023-10-12'),  
(8, 8, 8, 4, 'Nice and clean.', '2023-10-13'),  
(9, 9, 9, 5, 'Will come back!', '2023-10-14'),  
(10, 10, 10, 4, 'Great service.', '2023-10-15'),  
(11, 11, 11, 5, 'Fantastic views!', '2023-10-16'),  
(12, 12, 12, 3, 'Average stay.', '2023-10-17'),  
(13, 13, 13, 5, 'Luxurious!', '2023-10-18'),  
(14, 14, 14, 4, 'Very peaceful.', '2023-10-19'),  
(15, 15, 15, 5, 'Highly recommended!', '2023-10-20'),  
(16, 16, 16, 3, 'Needs improvement.', '2023-10-21'),  
(17, 17, 17, 4, 'Comfortable stay.', '2023-10-22'),  
(18, 18, 18, 5, 'Excellent!', '2023-10-23'),  
(19, 19, 19, 4, 'Great value.', '2023-10-24'),  
(20, 20, 20, 5, 'Wonderful experience!', '2023-10-25');
```

| id   | booking_id | reviewer_id | reviewee_id | rating | comment               | review_date |
|------|------------|-------------|-------------|--------|-----------------------|-------------|
| 1    | 1          | 1           | 1           | 5      | Great stay!           | 2023-10-05  |
| 2    | 2          | 2           | 2           | 4      | Loved the place!      | 2023-10-07  |
| 3    | 3          | 3           | 3           | 3      | It was okay.          | 2023-10-06  |
| 4    | 4          | 4           | 4           | 5      | Amazing experience!   | 2023-10-09  |
| 5    | 5          | 5           | 5           | 4      | Very comfortable.     | 2023-10-10  |
| 6    | 6          | 6           | 6           | 5      | Perfect location!     | 2023-10-11  |
| 7    | 7          | 7           | 7           | 3      | Could be better.      | 2023-10-12  |
| 8    | 8          | 8           | 8           | 4      | Nice and clean.       | 2023-10-13  |
| 9    | 9          | 9           | 9           | 5      | Will come back!       | 2023-10-14  |
| 10   | 10         | 10          | 10          | 4      | Great service.        | 2023-10-15  |
| 11   | 11         | 11          | 11          | 5      | Fantastic views!      | 2023-10-16  |
| 12   | 12         | 12          | 12          | 3      | Average stay.         | 2023-10-17  |
| 13   | 13         | 13          | 13          | 5      | Luxurious!            | 2023-10-18  |
| 14   | 14         | 14          | 14          | 4      | Very peaceful.        | 2023-10-19  |
| 15   | 15         | 15          | 15          | 5      | Highly recommended!   | 2023-10-20  |
| 16   | 16         | 16          | 16          | 3      | Needs improvement.    | 2023-10-21  |
| 17   | 17         | 17          | 17          | 4      | Comfortable stay.     | 2023-10-22  |
| 18   | 18         | 18          | 18          | 5      | Excellent!            | 2023-10-23  |
| 19   | 19         | 19          | 19          | 4      | Great value.          | 2023-10-24  |
| 20   | 20         | 20          | 20          | 5      | Wonderful experience! | 2023-10-25  |
| NULL | NULL       | NULL        | NULL        | NULL   | NULL                  | NULL        |

Amenity:  
This entity can store amenities available for properties.

Use the SQL statement to CREATE TABLE to define the structure of the Amenity table.

```
CREATE TABLE airbnb_company.Amenity (  
  id INT PRIMARY KEY AUTO_INCREMENT,  
  name VARCHAR(255) NOT NULL  
);
```

Inserting the name of amenities available on properties in the table using INSERT INTO statements.

```
INSERT INTO airbnb_company.Amenity (name)  
VALUES  
( 'Wi-Fi' ),  
( 'Pool' ),  
( 'Gym' ),  
( 'Parking' ),  
( 'Air Conditioning' ),  
( 'Heating' ),  
( 'Kitchen' ),  
( 'Washer' ),  
( 'Dryer' ),  
( 'TV' ),  
( 'Hot Tub' ),  
( 'Balcony' ),  
( 'Garden' ),  
( 'Terrace' ),  
( 'Fireplace' ),  
( 'Elevator' ),  
( 'Security System' ),  
( 'Pet Friendly' ),  
( 'Smoke Free' ),  
( 'Wheelchair Accessible' );
```

| id   | name                  |
|------|-----------------------|
| 1    | Wi-Fi                 |
| 2    | Pool                  |
| 3    | Gym                   |
| 4    | Parking               |
| 5    | Air Conditioning      |
| 6    | Heating               |
| 7    | Kitchen               |
| 8    | Washer                |
| 9    | Dryer                 |
| 10   | TV                    |
| 11   | Hot Tub               |
| 12   | Balcony               |
| 13   | Garden                |
| 14   | Terrace               |
| 15   | Fireplace             |
| 16   | Elevator              |
| 17   | Security System       |
| 18   | Pet Friendly          |
| 19   | Smoke Free            |
| 20   | Wheelchair Accessible |
| NULL | NULL                  |

## PropertyAmenity:

This entity can link properties to their amenities.

Use the SQL statement to CREATE TABLE to define the structure of the PropertyAmenity table.

```
• CREATE TABLE airbnb_company.PropertyAmenity (  
  property_id INT NOT NULL,  
  amenity_id INT NOT NULL,  
  PRIMARY KEY (property_id, amenity_id),  
  FOREIGN KEY (property_id) REFERENCES Property(id),  
  FOREIGN KEY (amenity_id) REFERENCES Amenity(id)  
);
```

Inserting the list of properties's amenities in the table using INSERT INTO statements.

```
• INSERT INTO airbnb_company.PropertyAmenity (property_id, amenity_id)  
VALUES  
(1, 1), (1, 2), (1, 3),  
(2, 4), (2, 5), (2, 6),  
(3, 7), (3, 8), (3, 9),  
(4, 10), (4, 11), (4, 12),  
(5, 13), (5, 14), (5, 15),  
(6, 16), (6, 17), (6, 18),  
(7, 19), (7, 20), (7, 1),  
(8, 2), (8, 3), (8, 4),  
(9, 5), (9, 6), (9, 7),  
(10, 8), (10, 9), (10, 10),  
(11, 11), (11, 12), (11, 13),  
(12, 14), (12, 15), (12, 16),  
(13, 17), (13, 18), (13, 19),  
(14, 20), (14, 1), (14, 2),  
(15, 3), (15, 4), (15, 5),  
(16, 6), (16, 7), (16, 8),  
(17, 9), (17, 10), (17, 11),  
(18, 12), (18, 13), (18, 14),  
(19, 15), (19, 16), (19, 17),  
(20, 18), (20, 19), (20, 20);
```

|  | property_id | amenity_id |
|--|-------------|------------|
|  | 1           | 1          |
|  | 7           | 1          |
|  | 14          | 1          |
|  | 1           | 2          |
|  | 8           | 2          |
|  | 14          | 2          |
|  | 1           | 3          |
|  | 8           | 3          |
|  | 15          | 3          |
|  | 2           | 4          |
|  | 8           | 4          |
|  | 15          | 4          |
|  | 2           | 5          |
|  | 9           | 5          |
|  | 15          | 5          |
|  | 2           | 6          |
|  | 9           | 6          |
|  | 16          | 6          |
|  | 3           | 7          |
|  | 9           | 7          |
|  | 16          | 7          |
|  | 3           | 8          |
|  | 10          | 8          |
|  | 16          | 8          |
|  | 3           | 9          |
|  | 10          | 9          |
|  | 17          | 9          |
|  | 4           | 10         |
|  | 10          | 10         |
|  | 17          | 10         |
|  | 4           | 11         |
|  | 11          | 11         |
|  | 17          | 11         |
|  | 4           | 12         |
|  | 11          | 12         |
|  | 18          | 12         |
|  | 5           | 13         |
|  | 11          | 13         |
|  | 18          | 13         |
|  | 5           | 14         |
|  | 12          | 14         |
|  | 18          | 14         |
|  | 5           | 15         |
|  | 12          | 15         |
|  | 19          | 15         |
|  | 6           | 16         |
|  | 12          | 16         |
|  | 19          | 16         |
|  | 6           | 17         |
|  | 13          | 17         |
|  | 19          | 17         |
|  | 6           | 18         |
|  | 13          | 18         |
|  | 20          | 18         |
|  | 7           | 19         |
|  | 13          | 19         |
|  | 20          | 19         |
|  | 7           | 20         |
|  | 14          | 20         |
|  | 20          | 20         |
|  | NULL        | NULL       |

## Availability:

This entity can track property availability dates and status.

Use the SQL statement to CREATE TABLE to define the structure of the Availability table.

```
• CREATE TABLE airbnb_company.Availability (  
  id INT PRIMARY KEY AUTO_INCREMENT,  
  property_id INT NOT NULL,  
  date DATE NOT NULL,  
  is_available BOOLEAN NOT NULL,  
  FOREIGN KEY (property_id) REFERENCES Property(id)  
);
```

Inserting the detailed information linked to the property availability dates and status in the table using INSERT INTO statements.

```
• INSERT INTO airbnb_company.Availability (property_id, date, is_available)  
VALUES  
(1, '2023-10-01', 0),  
(1, '2023-10-02', 0),  
(1, '2023-10-03', 0),  
(1, '2023-10-04', 0),  
(1, '2023-10-05', 0),  
(2, '2023-10-02', 0),  
(2, '2023-10-03', 0),  
(2, '2023-10-04', 0),  
(2, '2023-10-05', 0),  
(2, '2023-10-06', 0),  
(3, '2023-10-03', 0),  
(3, '2023-10-04', 0),  
(3, '2023-10-05', 0),  
(3, '2023-10-06', 0),  
(3, '2023-10-07', 0),  
(4, '2023-10-04', 0),  
(4, '2023-10-05', 0),  
(4, '2023-10-06', 0),  
(4, '2023-10-07', 0),  
(4, '2023-10-08', 0);
```

|  | id   | property_id | date       | is_available |
|--|------|-------------|------------|--------------|
|  | 1    | 1           | 2023-10-01 | 0            |
|  | 2    | 1           | 2023-10-02 | 0            |
|  | 3    | 1           | 2023-10-03 | 0            |
|  | 4    | 1           | 2023-10-04 | 0            |
|  | 5    | 1           | 2023-10-05 | 0            |
|  | 6    | 2           | 2023-10-02 | 0            |
|  | 7    | 2           | 2023-10-03 | 0            |
|  | 8    | 2           | 2023-10-04 | 0            |
|  | 9    | 2           | 2023-10-05 | 0            |
|  | 10   | 2           | 2023-10-06 | 0            |
|  | 11   | 3           | 2023-10-03 | 0            |
|  | 12   | 3           | 2023-10-04 | 0            |
|  | 13   | 3           | 2023-10-05 | 0            |
|  | 14   | 3           | 2023-10-06 | 0            |
|  | 15   | 3           | 2023-10-07 | 0            |
|  | 16   | 4           | 2023-10-04 | 0            |
|  | 17   | 4           | 2023-10-05 | 0            |
|  | 18   | 4           | 2023-10-06 | 0            |
|  | 19   | 4           | 2023-10-07 | 0            |
|  | 20   | 4           | 2023-10-08 | 0            |
|  | NULL | NULL        | NULL       | NULL         |

Commission:  
This entity can store commission details linked to booking.

Use the SQL statement to CREATE TABLE to define the structure of the Commission table.

```
CREATE TABLE airbnb_company.Commission (  
  id INT PRIMARY KEY AUTO_INCREMENT,  
  booking_id INT NOT NULL,  
  commission_rate DECIMAL(5, 2) NOT NULL,  
  commission_amount DECIMAL(10, 2) NOT NULL,  
  FOREIGN KEY (booking_id) REFERENCES Booking(id)  
);
```

Inserting the commission details linked to booking in the table using INSERT INTO statements.

```
INSERT INTO airbnb_company.Commission (booking_id, commission_rate, commission_amount)  
VALUES  
(1, 10.00, 40.00),  
(2, 10.00, 250.00),  
(3, 10.00, 60.00),  
(4, 10.00, 150.00),  
(5, 10.00, 75.00),  
(6, 10.00, 125.00),  
(7, 10.00, 60.00),  
(8, 10.00, 90.00),  
(9, 10.00, 45.00),  
(10, 10.00, 110.00),  
(11, 10.00, 200.00),  
(12, 10.00, 65.00),  
(13, 10.00, 300.00),  
(14, 10.00, 80.00),  
(15, 10.00, 225.00),  
(16, 10.00, 70.00),  
(17, 10.00, 85.00),  
(18, 10.00, 95.00),  
(19, 10.00, 175.00),  
(20, 10.00, 105.00);
```

| id   | booking_id | commission_rate | commission_amount |
|------|------------|-----------------|-------------------|
| 1    | 1          | 10.00           | 40.00             |
| 2    | 2          | 10.00           | 250.00            |
| 3    | 3          | 10.00           | 60.00             |
| 4    | 4          | 10.00           | 150.00            |
| 5    | 5          | 10.00           | 75.00             |
| 6    | 6          | 10.00           | 125.00            |
| 7    | 7          | 10.00           | 60.00             |
| 8    | 8          | 10.00           | 90.00             |
| 9    | 9          | 10.00           | 45.00             |
| 10   | 10         | 10.00           | 110.00            |
| 11   | 11         | 10.00           | 200.00            |
| 12   | 12         | 10.00           | 65.00             |
| 13   | 13         | 10.00           | 300.00            |
| 14   | 14         | 10.00           | 80.00             |
| 15   | 15         | 10.00           | 225.00            |
| 16   | 16         | 10.00           | 70.00             |
| 17   | 17         | 10.00           | 85.00             |
| 18   | 18         | 10.00           | 95.00             |
| 19   | 19         | 10.00           | 175.00            |
| 20   | 20         | 10.00           | 105.00            |
| NULL | NULL       | NULL            | NULL              |

Admin:  
This entity can store admin details linked to a user.

Use the SQL statement to CREATE TABLE to define the structure of the Admin table.

```
CREATE TABLE airbnb_company.Admin (  
  id INT PRIMARY KEY AUTO_INCREMENT,  
  name VARCHAR(255) NOT NULL,  
  email VARCHAR(255) UNIQUE NOT NULL,  
  user_id INT NOT NULL,  
  FOREIGN KEY (user_id) REFERENCES User(id)  
);
```

Inserting the detailed information of admin linked to user in the table using INSERT INTO statements.

```
INSERT INTO airbnb_company.Admin (name, email, user_id)  
VALUES  
( 'Alice Johnson', 'alice.johnson@gmail.com', 3),  
( 'Diana Evans', 'diana.evans@hotmail.com', 6),  
( 'George Clark', 'george.clark@hotmail.com', 9),  
( 'Jessica Hall', 'jessica.hall@hotmail.com', 12),  
( 'Michael King', 'michael.king@yahoo.com', 15),  
( 'Paula Turner', 'paula.turner@hotmail.com', 18);
```

|  | id   | name          | email                    | user_id |
|--|------|---------------|--------------------------|---------|
|  | 1    | Alice Johnson | alice.johnson@gmail.com  | 3       |
|  | 2    | Diana Evans   | diana.evans@hotmail.com  | 6       |
|  | 3    | George Clark  | george.clark@hotmail.com | 9       |
|  | 4    | Jessica Hall  | jessica.hall@hotmail.com | 12      |
|  | 5    | Michael King  | michael.king@yahoo.com   | 15      |
|  | 6    | Paula Turner  | paula.turner@hotmail.com | 18      |
|  | NULL | NULL          | NULL                     | NULL    |
|  |      |               |                          |         |

## Payout:

This entity can store payout details to host, including amount and approval status.

Use the SQL statement to CREATE TABLE to define the structure of the Payout table.

```
CREATE TABLE airbnb_company.Payout (  
  id INT PRIMARY KEY AUTO_INCREMENT,  
  host_id INT NOT NULL,  
  amount DECIMAL(10, 2) NOT NULL,  
  payout_date DATE NOT NULL,  
  approved_by INT,  
  FOREIGN KEY (host_id) REFERENCES Host(id),  
  FOREIGN KEY (approved_by) REFERENCES Admin(id)  
);
```

Inserting the payout details information linked to host and admin in the table using INSERT INTO statements.

```
INSERT INTO airbnb_company.Payout (host_id, amount, payout_date, approved_by)  
VALUES  
(1, 360.00, '2023-10-06', 3),  
(2, 2250.00, '2023-10-08', 3),  
(3, 540.00, '2023-10-07', 3),  
(4, 1350.00, '2023-10-10', 3),  
(5, 675.00, '2023-10-11', 3),  
(6, 1125.00, '2023-10-12', 3),  
(7, 540.00, '2023-10-13', 3),  
(8, 810.00, '2023-10-14', 3),  
(9, 405.00, '2023-10-15', 3),  
(10, 990.00, '2023-10-16', 3),  
(11, 1800.00, '2023-10-17', 3),  
(12, 585.00, '2023-10-18', 3),  
(13, 2700.00, '2023-10-19', 3),  
(14, 720.00, '2023-10-20', 3),  
(15, 2025.00, '2023-10-21', 3),  
(16, 630.00, '2023-10-22', 3),  
(17, 765.00, '2023-10-23', 3),  
(18, 855.00, '2023-10-24', 3),  
(19, 1575.00, '2023-10-25', 3),  
(20, 945.00, '2023-10-26', 3);
```

|  | id   | host_id | amount  | payout_date | approved_by |
|--|------|---------|---------|-------------|-------------|
|  | 1    | 1       | 360.00  | 2023-10-06  | 3           |
|  | 2    | 2       | 2250.00 | 2023-10-08  | 3           |
|  | 3    | 3       | 540.00  | 2023-10-07  | 3           |
|  | 4    | 4       | 1350.00 | 2023-10-10  | 3           |
|  | 5    | 5       | 675.00  | 2023-10-11  | 3           |
|  | 6    | 6       | 1125.00 | 2023-10-12  | 3           |
|  | 7    | 7       | 540.00  | 2023-10-13  | 3           |
|  | 8    | 8       | 810.00  | 2023-10-14  | 3           |
|  | 9    | 9       | 405.00  | 2023-10-15  | 3           |
|  | 10   | 10      | 990.00  | 2023-10-16  | 3           |
|  | 11   | 11      | 1800.00 | 2023-10-17  | 3           |
|  | 12   | 12      | 585.00  | 2023-10-18  | 3           |
|  | 13   | 13      | 2700.00 | 2023-10-19  | 3           |
|  | 14   | 14      | 720.00  | 2023-10-20  | 3           |
|  | 15   | 15      | 2025.00 | 2023-10-21  | 3           |
|  | 16   | 16      | 630.00  | 2023-10-22  | 3           |
|  | 17   | 17      | 765.00  | 2023-10-23  | 3           |
|  | 18   | 18      | 855.00  | 2023-10-24  | 3           |
|  | 19   | 19      | 1575.00 | 2023-10-25  | 3           |
|  | 20   | 20      | 945.00  | 2023-10-26  | 3           |
|  | NULL | NULL    | NULL    | NULL        | NULL        |
|  |      |         |         |             |             |

Message:  
This entity can store messages between users with sender, receiver, and timestamp.

Use the SQL statement to CREATE TABLE to define the structure of the Message table.

```
CREATE TABLE airbnb_company.Message (  
  id INT PRIMARY KEY AUTO_INCREMENT,  
  sender_id INT NOT NULL,  
  receiver_id INT NOT NULL,  
  message_text TEXT NOT NULL,  
  timestamp DATETIME NOT NULL,  
  FOREIGN KEY (sender_id) REFERENCES User(id),  
  FOREIGN KEY (receiver_id) REFERENCES User(id)  
);
```

Inserting the messages linked to user with sender, receiver and timestamp in the table using INSERT INTO statements.

```
INSERT INTO airbnb_company.Message (sender_id, receiver_id, message_text, timestamp)  
VALUES  
(1, 2, 'Hi, can I check in early?', '2023-10-01 10:00:00'),  
(2, 1, 'Sure, what time?', '2023-10-01 10:05:00'),  
(3, 4, 'Is the pool heated?', '2023-10-02 11:00:00'),  
(4, 3, 'Yes, it is.', '2023-10-02 11:10:00'),  
(5, 6, 'Do you have parking?', '2023-10-03 12:00:00'),  
(6, 5, 'Yes, we do.', '2023-10-03 12:15:00'),  
(7, 8, 'Is the kitchen fully equipped?', '2023-10-04 13:00:00'),  
(8, 7, 'Yes, it has everything you need.', '2023-10-04 13:20:00'),  
(9, 10, 'Can I bring my pet?', '2023-10-05 14:00:00'),  
(10, 9, 'Yes, pets are allowed.', '2023-10-05 14:30:00'),  
(11, 12, 'Is there a gym nearby?', '2023-10-06 15:00:00'),  
(12, 11, 'Yes, just a block away.', '2023-10-06 15:10:00'),  
(13, 14, 'Do you have a crib?', '2023-10-07 16:00:00'),  
(14, 13, 'Yes, we can provide one.', '2023-10-07 16:20:00'),  
(15, 16, 'Is the balcony safe for kids?', '2023-10-08 17:00:00'),  
(16, 15, 'Yes, it has a safety rail.', '2023-10-08 17:15:00'),  
(17, 18, 'Can I get a late checkout?', '2023-10-09 18:00:00'),  
(18, 17, 'Sure, until 2 PM.', '2023-10-09 18:30:00'),  
(19, 20, 'Is there a supermarket nearby?', '2023-10-10 19:00:00'),  
(20, 19, 'Yes, just around the corner.', '2023-10-10 19:10:00');
```

|  | id   | sender_id | receiver_id | message_text                     | timestamp           |
|--|------|-----------|-------------|----------------------------------|---------------------|
|  | 1    | 1         | 2           | Hi, can I check in early?        | 2023-10-01 10:00:00 |
|  | 2    | 2         | 1           | Sure, what time?                 | 2023-10-01 10:05:00 |
|  | 3    | 3         | 4           | Is the pool heated?              | 2023-10-02 11:00:00 |
|  | 4    | 4         | 3           | Yes, it is.                      | 2023-10-02 11:10:00 |
|  | 5    | 5         | 6           | Do you have parking?             | 2023-10-03 12:00:00 |
|  | 6    | 6         | 5           | Yes, we do.                      | 2023-10-03 12:15:00 |
|  | 7    | 7         | 8           | Is the kitchen fully equipped?   | 2023-10-04 13:00:00 |
|  | 8    | 8         | 7           | Yes, it has everything you need. | 2023-10-04 13:20:00 |
|  | 9    | 9         | 10          | Can I bring my pet?              | 2023-10-05 14:00:00 |
|  | 10   | 10        | 9           | Yes, pets are allowed.           | 2023-10-05 14:30:00 |
|  | 11   | 11        | 12          | Is there a gym nearby?           | 2023-10-06 15:00:00 |
|  | 12   | 12        | 11          | Yes, just a block away.          | 2023-10-06 15:10:00 |
|  | 13   | 13        | 14          | Do you have a crib?              | 2023-10-07 16:00:00 |
|  | 14   | 14        | 13          | Yes, we can provide one.         | 2023-10-07 16:20:00 |
|  | 15   | 15        | 16          | Is the balcony safe for kids?    | 2023-10-08 17:00:00 |
|  | 16   | 16        | 15          | Yes, it has a safety rail.       | 2023-10-08 17:15:00 |
|  | 17   | 17        | 18          | Can I get a late checkout?       | 2023-10-09 18:00:00 |
|  | 18   | 18        | 17          | Sure, until 2 PM.                | 2023-10-09 18:30:00 |
|  | 19   | 19        | 20          | Is there a supermarket nearby?   | 2023-10-10 19:00:00 |
|  | 20   | 20        | 19          | Yes, just around the corner.     | 2023-10-10 19:10:00 |
|  | NULL | NULL      | NULL        | NULL                             | NULL                |

SocialMedia:

This entity can store the user social media profiles and links.

Use the SQL statement to CREATE TABLE to define the structure of the SocialMedia table.

```
• CREATE TABLE airbnb_company.SocialMedia (  
  id INT PRIMARY KEY AUTO_INCREMENT,  
  user_id INT NOT NULL,  
  platform_name VARCHAR(255) NOT NULL,  
  profile_link VARCHAR(255) NOT NULL,  
  FOREIGN KEY (user_id) REFERENCES User(id)  
);  
  
• INSERT INTO airbnb_company.SocialMedia (user_id, platform_name, profile_link)  
VALUES  
(1, 'Facebook', 'https://facebook.com/johndoe'),  
(2, 'Instagram', 'https://instagram.com/janesmith'),  
(3, 'Twitter', 'https://twitter.com/alicejohnson'),  
(4, 'LinkedIn', 'https://linkedin.com/in/bobbrown'),  
(5, 'Facebook', 'https://facebook.com/charliedavis'),  
(6, 'Instagram', 'https://instagram.com/dianaevans'),  
(7, 'Twitter', 'https://twitter.com/ethangreen'),  
(8, 'LinkedIn', 'https://linkedin.com/in/fionaharris'),  
(9, 'Facebook', 'https://facebook.com/georgeclark'),  
(10, 'Instagram', 'https://instagram.com/hannahlewis'),  
(11, 'Twitter', 'https://twitter.com/ianwalker'),  
(12, 'LinkedIn', 'https://linkedin.com/in/jessicahall'),  
(13, 'Facebook', 'https://facebook.com/kevinallen'),  
(14, 'Instagram', 'https://instagram.com/laurayoung'),  
(15, 'Twitter', 'https://twitter.com/michaelking'),  
(16, 'LinkedIn', 'https://linkedin.com/in/ninawright'),  
(17, 'Facebook', 'https://facebook.com/oscarscott'),  
(18, 'Instagram', 'https://instagram.com/paulaturner'),  
(19, 'Twitter', 'https://twitter.com/quincyadams'),  
(20, 'LinkedIn', 'https://linkedin.com/in/rachelwhite');
```

| id   | user_id | platform_name | profile_link                        |
|------|---------|---------------|-------------------------------------|
| 1    | 1       | Facebook      | https://facebook.com/johndoe        |
| 2    | 2       | Instagram     | https://instagram.com/janesmith     |
| 3    | 3       | Twitter       | https://twitter.com/alicejohnson    |
| 4    | 4       | LinkedIn      | https://linkedin.com/in/bobbrown    |
| 5    | 5       | Facebook      | https://facebook.com/charliedavis   |
| 6    | 6       | Instagram     | https://instagram.com/dianaevans    |
| 7    | 7       | Twitter       | https://twitter.com/ethangreen      |
| 8    | 8       | LinkedIn      | https://linkedin.com/in/fionaharris |
| 9    | 9       | Facebook      | https://facebook.com/georgeclark    |
| 10   | 10      | Instagram     | https://instagram.com/hannahlewis   |
| 11   | 11      | Twitter       | https://twitter.com/ianwalker       |
| 12   | 12      | LinkedIn      | https://linkedin.com/in/jessicahall |
| 13   | 13      | Facebook      | https://facebook.com/kevinallen     |
| 14   | 14      | Instagram     | https://instagram.com/laurayoung    |
| 15   | 15      | Twitter       | https://twitter.com/michaelking     |
| 16   | 16      | LinkedIn      | https://linkedin.com/in/ninawright  |
| 17   | 17      | Facebook      | https://facebook.com/oscarscott     |
| 18   | 18      | Instagram     | https://instagram.com/paulaturner   |
| 19   | 19      | Twitter       | https://twitter.com/quincyadams     |
| 20   | 20      | LinkedIn      | https://linkedin.com/in/rachelwhite |
| NULL | NULL    | NULL          | NULL                                |

Dispute:

This entity can store disputes linked to bookings, including complaint text and resolution status.

Use the SQL statement to CREATE TABLE to define the structure of the Dispute table.

```
CREATE TABLE airbnb_company.Dispute (  
  id INT PRIMARY KEY AUTO_INCREMENT,  
  booking_id INT NOT NULL,  
  complaint_text TEXT NOT NULL,  
  resolved_by INT,  
  resolution_status ENUM('Open', 'Resolved', 'Closed') NOT NULL,  
  FOREIGN KEY (booking_id) REFERENCES Booking(id),  
  FOREIGN KEY (resolved_by) REFERENCES Admin(id)  
);
```

Inserting the disputes linked to booking and admin, including the complaint text and resolution status (Open , Resolved, Closed) in the table using INSERT INTO statements.

```
INSERT INTO airbnb_company.Dispute (booking_id, complaint_text, resolved_by, resolution_status)  
VALUES  
(1, 'The apartment was not clean.', 3, 'Resolved'),  
(2, 'The pool was not heated.', 3, 'Open'),  
(3, 'The kitchen was not fully equipped.', 3, 'Closed'),  
(4, 'The parking space was not available.', 3, 'Resolved'),  
(5, 'The Wi-Fi was not working.', 3, 'Open'),  
(6, 'The TV was not functioning.', 3, 'Closed'),  
(7, 'The air conditioning was not working.', 3, 'Resolved'),  
(8, 'The check-in process was delayed.', 3, 'Open'),  
(9, 'The bathroom was not clean.', 3, 'Closed'),  
(10, 'The balcony door was broken.', 3, 'Resolved'),  
(11, 'The hot tub was not working.', 3, 'Open'),  
(12, 'The fireplace was not functional.', 3, 'Closed'),  
(13, 'The elevator was out of service.', 3, 'Resolved'),  
(14, 'The security system was not working.', 3, 'Open'),  
(15, 'The pet policy was not clear.', 3, 'Closed'),  
(16, 'The smoke alarm was beeping.', 3, 'Resolved'),  
(17, 'The wheelchair access was not as described.', 3, 'Open'),  
(18, 'The garden was not maintained.', 3, 'Closed'),  
(19, 'The terrace was not safe.', 3, 'Resolved'),  
(20, 'The check-out process was delayed.', 3, 'Open');
```

|  | id   | booking_id | complaint_text                              | resolved_by | resolution_status |
|--|------|------------|---------------------------------------------|-------------|-------------------|
|  | 1    | 1          | The apartment was not clean.                | 3           | Resolved          |
|  | 2    | 2          | The pool was not heated.                    | 3           | Open              |
|  | 3    | 3          | The kitchen was not fully equipped.         | 3           | Closed            |
|  | 4    | 4          | The parking space was not available.        | 3           | Resolved          |
|  | 5    | 5          | The Wi-Fi was not working.                  | 3           | Open              |
|  | 6    | 6          | The TV was not functioning.                 | 3           | Closed            |
|  | 7    | 7          | The air conditioning was not working.       | 3           | Resolved          |
|  | 8    | 8          | The check-in process was delayed.           | 3           | Open              |
|  | 9    | 9          | The bathroom was not clean.                 | 3           | Closed            |
|  | 10   | 10         | The balcony door was broken.                | 3           | Resolved          |
|  | 11   | 11         | The hot tub was not working.                | 3           | Open              |
|  | 12   | 12         | The fireplace was not functional.           | 3           | Closed            |
|  | 13   | 13         | The elevator was out of service.            | 3           | Resolved          |
|  | 14   | 14         | The security system was not working.        | 3           | Open              |
|  | 15   | 15         | The pet policy was not clear.               | 3           | Closed            |
|  | 16   | 16         | The smoke alarm was beeping.                | 3           | Resolved          |
|  | 17   | 17         | The wheelchair access was not as described. | 3           | Open              |
|  | 18   | 18         | The garden was not maintained.              | 3           | Closed            |
|  | 19   | 19         | The terrace was not safe.                   | 3           | Resolved          |
|  | 20   | 20         | The check-out process was delayed.          | 3           | Open              |
|  | NULL | NULL       | NULL                                        | NULL        | NULL              |
|  |      |            |                                             |             |                   |

Notification:

This entity can store user notifications with message and timestamp.

Use the SQL statement to CREATE TABLE to define the structure of the Notification table.

```
CREATE TABLE airbnb_company.Notification (  
  id INT PRIMARY KEY AUTO_INCREMENT,  
  user_id INT NOT NULL,  
  message TEXT NOT NULL,  
  timestamp DATETIME NOT NULL,  
  FOREIGN KEY (user_id) REFERENCES User(id)  
);
```

Inserting the user notification with message and timestamp in the table using INSERT INTO statements.

```
INSERT INTO airbnb_company.Notification (user_id, message, timestamp)  
VALUES  
(1, 'Your booking is confirmed.', '2023-10-01 10:00:00'),  
(2, 'Your booking is confirmed.', '2023-10-02 11:00:00'),  
(3, 'Your booking is confirmed.', '2023-10-03 12:00:00'),  
(4, 'Your booking is confirmed.', '2023-10-04 13:00:00'),  
(5, 'Your booking is confirmed.', '2023-10-05 14:00:00'),  
(6, 'Your booking is confirmed.', '2023-10-06 15:00:00'),  
(7, 'Your booking is confirmed.', '2023-10-07 16:00:00'),  
(8, 'Your booking is confirmed.', '2023-10-08 17:00:00'),  
(9, 'Your booking is confirmed.', '2023-10-09 18:00:00'),  
(10, 'Your booking is confirmed.', '2023-10-10 19:00:00'),  
(11, 'Your booking is confirmed.', '2023-10-11 20:00:00'),  
(12, 'Your booking is confirmed.', '2023-10-12 21:00:00'),  
(13, 'Your booking is confirmed.', '2023-10-13 22:00:00'),  
(14, 'Your booking is confirmed.', '2023-10-14 23:00:00'),  
(15, 'Your booking is confirmed.', '2023-10-15 10:00:00'),  
(16, 'Your booking is confirmed.', '2023-10-16 11:00:00'),  
(17, 'Your booking is confirmed.', '2023-10-17 12:00:00'),  
(18, 'Your booking is confirmed.', '2023-10-18 13:00:00'),  
(19, 'Your booking is confirmed.', '2023-10-19 14:00:00'),  
(20, 'Your booking is confirmed.', '2023-10-20 15:00:00');
```

| id   | user_id | message                    | timestamp           |
|------|---------|----------------------------|---------------------|
| 1    | 1       | Your booking is confirmed. | 2023-10-01 10:00:00 |
| 2    | 2       | Your booking is confirmed. | 2023-10-02 11:00:00 |
| 3    | 3       | Your booking is confirmed. | 2023-10-03 12:00:00 |
| 4    | 4       | Your booking is confirmed. | 2023-10-04 13:00:00 |
| 5    | 5       | Your booking is confirmed. | 2023-10-05 14:00:00 |
| 6    | 6       | Your booking is confirmed. | 2023-10-06 15:00:00 |
| 7    | 7       | Your booking is confirmed. | 2023-10-07 16:00:00 |
| 8    | 8       | Your booking is confirmed. | 2023-10-08 17:00:00 |
| 9    | 9       | Your booking is confirmed. | 2023-10-09 18:00:00 |
| 10   | 10      | Your booking is confirmed. | 2023-10-10 19:00:00 |
| 11   | 11      | Your booking is confirmed. | 2023-10-11 20:00:00 |
| 12   | 12      | Your booking is confirmed. | 2023-10-12 21:00:00 |
| 13   | 13      | Your booking is confirmed. | 2023-10-13 22:00:00 |
| 14   | 14      | Your booking is confirmed. | 2023-10-14 23:00:00 |
| 15   | 15      | Your booking is confirmed. | 2023-10-15 10:00:00 |
| 16   | 16      | Your booking is confirmed. | 2023-10-16 11:00:00 |
| 17   | 17      | Your booking is confirmed. | 2023-10-17 12:00:00 |
| 18   | 18      | Your booking is confirmed. | 2023-10-18 13:00:00 |
| 19   | 19      | Your booking is confirmed. | 2023-10-19 14:00:00 |
| 20   | 20      | Your booking is confirmed. | 2023-10-20 15:00:00 |
| NULL | NULL    | NULL                       | NULL                |

Income:

This entity can store user income details including source, amount, and date.

Use the SQL statement to CREATE TABLE to define the structure of the Income table.

```
• CREATE TABLE airbnb_company.Income (  
    id INT PRIMARY KEY AUTO_INCREMENT,  
    user_id INT NOT NULL,  
    source VARCHAR(255) NOT NULL,  
    amount DECIMAL(10, 2) NOT NULL,  
    income_date DATE NOT NULL,  
    FOREIGN KEY (user_id) REFERENCES User(id)  
);
```

Inserting the detailed income information linked to user including source, amount, and date in the table using INSERT INTO statements.

```
• INSERT INTO airbnb_company.Income (user_id, source, amount, income_date)  
VALUES  
(1, 'Property Rental', 1000.00, '2023-10-01'),  
(2, 'Property Rental', 2500.00, '2023-10-02'),  
(3, 'Property Rental', 600.00, '2023-10-03'),  
(4, 'Property Rental', 1500.00, '2023-10-04'),  
(5, 'Property Rental', 750.00, '2023-10-05'),  
(6, 'Property Rental', 1250.00, '2023-10-06'),  
(7, 'Property Rental', 600.00, '2023-10-07'),  
(8, 'Property Rental', 900.00, '2023-10-08'),  
(9, 'Property Rental', 450.00, '2023-10-09'),  
(10, 'Property Rental', 1100.00, '2023-10-10'),  
(11, 'Property Rental', 2000.00, '2023-10-11'),  
(12, 'Property Rental', 650.00, '2023-10-12'),  
(13, 'Property Rental', 3000.00, '2023-10-13'),  
(14, 'Property Rental', 800.00, '2023-10-14'),  
(15, 'Property Rental', 2250.00, '2023-10-15'),  
(16, 'Property Rental', 700.00, '2023-10-16'),  
(17, 'Property Rental', 850.00, '2023-10-17'),  
(18, 'Property Rental', 950.00, '2023-10-18'),  
(19, 'Property Rental', 1750.00, '2023-10-19'),  
(20, 'Property Rental', 1050.00, '2023-10-20');
```

|  | id   | user_id | source          | amount  | income_date |
|--|------|---------|-----------------|---------|-------------|
|  | 1    | 1       | Property Rental | 1000.00 | 2023-10-01  |
|  | 2    | 2       | Property Rental | 2500.00 | 2023-10-02  |
|  | 3    | 3       | Property Rental | 600.00  | 2023-10-03  |
|  | 4    | 4       | Property Rental | 1500.00 | 2023-10-04  |
|  | 5    | 5       | Property Rental | 750.00  | 2023-10-05  |
|  | 6    | 6       | Property Rental | 1250.00 | 2023-10-06  |
|  | 7    | 7       | Property Rental | 600.00  | 2023-10-07  |
|  | 8    | 8       | Property Rental | 900.00  | 2023-10-08  |
|  | 9    | 9       | Property Rental | 450.00  | 2023-10-09  |
|  | 10   | 10      | Property Rental | 1100.00 | 2023-10-10  |
|  | 11   | 11      | Property Rental | 2000.00 | 2023-10-11  |
|  | 12   | 12      | Property Rental | 650.00  | 2023-10-12  |
|  | 13   | 13      | Property Rental | 3000.00 | 2023-10-13  |
|  | 14   | 14      | Property Rental | 800.00  | 2023-10-14  |
|  | 15   | 15      | Property Rental | 2250.00 | 2023-10-15  |
|  | 16   | 16      | Property Rental | 700.00  | 2023-10-16  |
|  | 17   | 17      | Property Rental | 850.00  | 2023-10-17  |
|  | 18   | 18      | Property Rental | 950.00  | 2023-10-18  |
|  | 19   | 19      | Property Rental | 1750.00 | 2023-10-19  |
|  | 20   | 20      | Property Rental | 1050.00 | 2023-10-20  |
|  | NULL | NULL    | NULL            | NULL    | NULL        |

Language:

This entity can store the language options:

Use the SQL statement to CREATE TABLE to define the structure of the Language table.

```
• CREATE TABLE airbnb_company.Language (  
    id INT PRIMARY KEY AUTO_INCREMENT,  
    name VARCHAR(255) NOT NULL  
);
```

Inserting the language options in the table using INSERT INTO statements.

```
• INSERT INTO airbnb_company.Language (name)  
VALUES  
( 'English' ),  
( 'Spanish' ),  
( 'French' ),  
( 'German' ),  
( 'Italian' ),  
( 'Portuguese' ),  
( 'Russian' ),  
( 'Chinese' ),  
( 'Japanese' ),  
( 'Korean' ),  
( 'Arabic' ),  
( 'Hindi' ),  
( 'Turkish' ),  
( 'Dutch' ),  
( 'Swedish' ),  
( 'Danish' ),  
( 'Norwegian' ),  
( 'Finnish' ),  
( 'Polish' ),  
( 'Greek' );
```

|  | id   | name       |
|--|------|------------|
|  | 1    | English    |
|  | 2    | Spanish    |
|  | 3    | French     |
|  | 4    | German     |
|  | 5    | Italian    |
|  | 6    | Portuguese |
|  | 7    | Russian    |
|  | 8    | Chinese    |
|  | 9    | Japanese   |
|  | 10   | Korean     |
|  | 11   | Arabic     |
|  | 12   | Hindi      |
|  | 13   | Turkish    |
|  | 14   | Dutch      |
|  | 15   | Swedish    |
|  | 16   | Danish     |
|  | 17   | Norwegian  |
|  | 18   | Finnish    |
|  | 19   | Polish     |
|  | 20   | Greek      |
|  | NULL | NULL       |

## UserLanguage:

This entity can link users to their spoken languages.

Use the SQL statement to CREATE TABLE to define the structure of the UserLanguage table.

```
CREATE TABLE airbnb_company.UserLanguage (  
  user_id INT NOT NULL,  
  language_id INT NOT NULL,  
  PRIMARY KEY (user_id, language_id),  
  FOREIGN KEY (user_id) REFERENCES User(id),  
  FOREIGN KEY (language_id) REFERENCES Language(id)  
);
```

Inserting the spoken languages links to users in the table using INSERT INTO statements.

```
INSERT INTO airbnb_company.UserLanguage (user_id, language_id)  
VALUES  
(1, 1), (1, 2),  
(2, 3), (2, 4),  
(3, 5), (3, 6),  
(4, 7), (4, 8),  
(5, 9), (5, 10),  
(6, 11), (6, 12),  
(7, 13), (7, 14),  
(8, 15), (8, 16),  
(9, 17), (9, 18),  
(10, 19), (10, 20),  
(11, 1), (11, 3),  
(12, 2), (12, 4),  
(13, 5), (13, 7),  
(14, 6), (14, 8),  
(15, 9), (15, 11),  
(16, 10), (16, 12),  
(17, 13), (17, 15),  
(18, 14), (18, 16),  
(19, 17), (19, 19),  
(20, 18), (20, 20);
```

|  | user_id | language_id |
|--|---------|-------------|
|  | 1       | 1           |
|  | 11      | 1           |
|  | 1       | 2           |
|  | 12      | 2           |
|  | 2       | 3           |
|  | 11      | 3           |
|  | 2       | 4           |
|  | 12      | 4           |
|  | 3       | 5           |
|  | 13      | 5           |
|  | 3       | 6           |
|  | 14      | 6           |
|  | 4       | 7           |
|  | 13      | 7           |
|  | 4       | 8           |
|  | 14      | 8           |
|  | 5       | 9           |
|  | 15      | 9           |
|  | 5       | 10          |
|  | 16      | 10          |
|  | 6       | 11          |
|  | 15      | 11          |
|  | 6       | 12          |
|  | 16      | 12          |
|  | 7       | 13          |
|  | 17      | 13          |
|  | 7       | 14          |
|  | 18      | 14          |
|  | 8       | 15          |
|  | 17      | 15          |
|  | 8       | 16          |
|  | 18      | 16          |
|  | 9       | 17          |
|  | 19      | 17          |
|  | 9       | 18          |
|  | 20      | 18          |
|  | 10      | 19          |
|  | 19      | 19          |
|  | 10      | 20          |
|  | 20      | 20          |
|  | NULL    | NULL        |